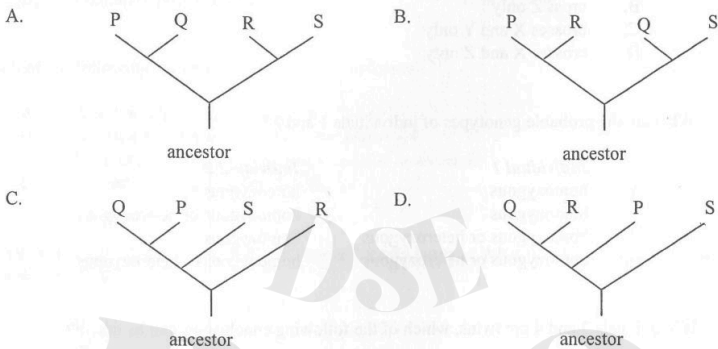


1. 2013Q20

Questions 20 refer to the nucleotide sequence of a certain functional gene segment found in four different species of organisms P, Q, R and S:

A A C G T C G A A A (organism P)
A A C C T C G A A A (organism Q)
A G G C T A G A A A (organism R)
A G G C T A G T A A (organism S)

Based on the above information, which of the following diagrams best represents the evolutionary tree of organisms P, Q, R and S?



2. 2013Q22

Which of the following observations is **not** related to the process of evolution?

A. Some variations are not inheritable.
B. Organisms compete for resources to survive.
C. Environmental changes give stress to organisms.
D. Variations exist among different individuals of the same species.

3. 2014Q13

The average height of men in a developed country rose by 10 cm between 1890 and 1980. Which of the following is the most probable reason for this observation?

A. Chemical pollution induced mutations leading to the increase in height.
B. Spontaneous mutation resulted in a shift to taller height.
C. The better nutrition supply promoted growth.
D. A taller height had a better chance of survival.

4. 2014Q18

Question 18 refers to the following photographs of two different fish:



Which of the following allow further study of the phylogenetic relationship between the two fish?

(1) Compare the amino acid sequences of their functional proteins
(2) Compare their internal body structure
(3) Compare their living habitats and behaviours

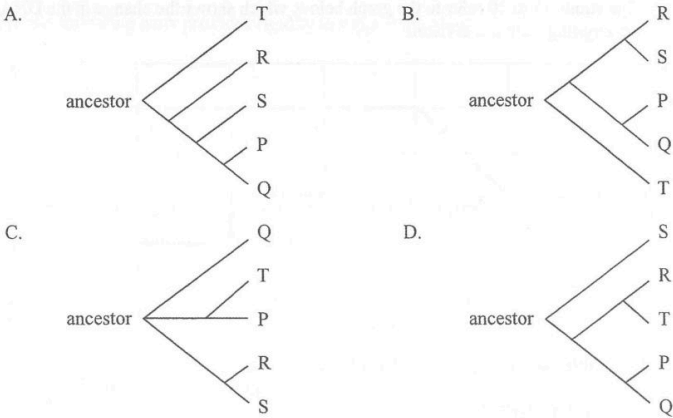
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

5. 2016Q14

Question 14 refers to the information below. Five new species of eubacteria were discovered in Antarctic ice core samples. The nucleotide sequences of the gene that codes for an essential protein of these new species were compared. The table below shows the number of nucleotide differences between the species:

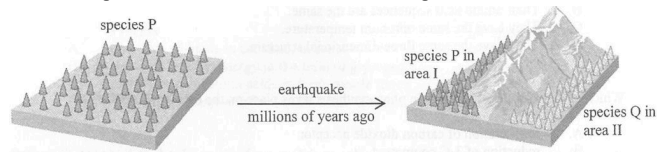
Species	Number of nucleotide differences				
	P	Q	R	S	T
P	-	4	12	11	22
Q		-	12	11	19
R			-	4	22
S				-	22
T					-

Which of the following evolutionary trees best illustrates the phylogenetic relationship of the five species



6. 2017Q10

Questions 10 and 11 refer to the diagram below. A high mountain resulting from an earthquake millions of years ago has led to the separation of areas I and II. A new tree species Q is found in area II.



Which of the following processes were likely to have been involved in the formation of new species Q?

- (1) mutation
(2) isolation
(3) natural selection
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

7. 2017Q11

Which of the following descriptions of the above incident is most likely to be correct?

- A. Q is more adaptive than P.
B. P grows equally well in areas I and II.
C. Areas I and II have similar environmental conditions.
D. P and Q belong to the same Family in the classification system.

8. 2020Q15

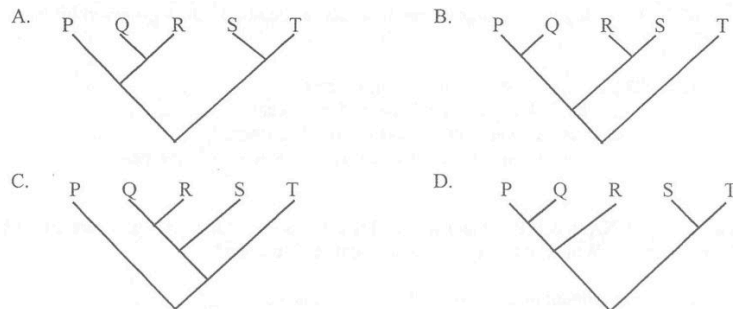
The table below shows the presence or absence of some traits in five species:

Species	Trait				
	1	2	3	4	5
P	+	+	-	+	-
Q	+	+	-	-	-
R	+	-	-	-	+
S	+	-	+	-	+
T	-	-	-	-	-

Key:

- + presence of trait
- absence of trait

Which of the following evolutionary trees best illustrates the phylogenetic relationship of the five species?



9. 2020Q16

Australia has some plant and animal species which cannot be found elsewhere in the world. Which of the following is likely to be the key factor that led to the occurrence of these unique plant and animal species in Australia?

- A. isolation
B. mutation
C. competition
D. environmental stress

10. 2020Q21

In flowering plants, asexual reproduction is considered less favourable than sexual reproduction in terms of natural selection because offspring produced from asexual reproduction

- A. are genetically identical to each other.
B. have keen competition with each other.
C. cannot invade and colonise new environments.
D. can grow rapidly only in favourable conditions.

11. 2022Q7

The table below shows the comparison among three groups of organisms X, Y and Z:

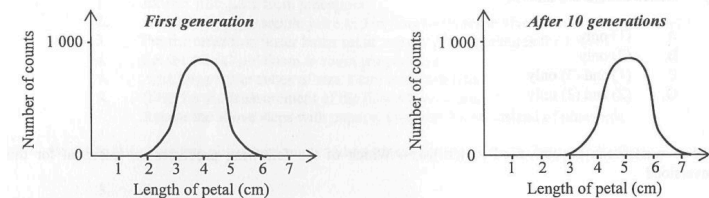
	X	Y	Z
Cell wall	Present	Present	Absent
Similarity of the amino acid sequence of a protein found in all three organisms (X as a reference)	100%	45%	55%

With reference to the information above, which of the following can be deduced?

- A. Both X and Y are prokaryotes.
B. Y is least advanced among the three.
C. X has a closer phylogenetic relationship with Z than Y.
D. Y has a closer phylogenetic relationship with Z than X.

12. 2023Q25

Question 25 refers to the information about the introduction of a foreign plant species to a local botanic garden. Over the years, the gardener noted that there was a change in the size of its flowers. The graphs below show the variations of the petal length of this plant species in the first generation and after 10 generations.



Which of the following is the most probable explanation for the change in the petal length of this plant species?

- A. Larger flowers produce more nectar.
B. Local climate favours the growth of this plant species.
C. Local insects are attracted by flowers with longer petals.
D. Mutations occur and accumulate throughout the 10 generations.

13. 2023Q29

Question 29 refers to the phylogenetic relationship of five bacterial species. Enzyme X is found in most bacteria. The table below shows the number of nucleotide differences in the genes encoding enzyme X of five bacterial species (J, K, L, M, and N).

	J	K	L	M	N
J	0	61	32	112	89
K	-	0	50	119	94
L	-	-	0	119	77
M	-	-	-	0	124
N	-	-	-	-	0

With regard to the phylogenetic relationship to L, arrange the species in the correct order from closely related to distantly related.

A. L, J, K, N, M

B. L, J, M, N, K

C. L, M, N, K, J

D. L, N, M, K, J

14. 2024Q11

Sharks are considered ‘living fossils’ because their appearance closely resembles shark fossils from millions of years ago. Which of the following is the most likely reason for the close resemblance?

A. Sharks have few predators.

B. Sharks have a wide range of prey.

C. A shark’s body is well-adapted to the ocean environment.

D. A shark’s body is most advanced as compared to other species in oceans.

15. 2024Q12

Questions 12 and 13 refer to the diagram below, which shows the phylogenetic relationships between humans and some primates:

Which of the following statements about these primates is correct?

A. Pileated gibbons share a common ancestor with gorillas and humans.

B. Gorillas have a closer relationship with ring tailed lemurs than white face sakis.

C. White face sakis are the common ancestor of tufted capuchins and white-fronted capuchins.

D. The difference between gorillas and humans is the same as that of ring-tailed lemurs and greater bamboo lemurs.

16. 2024Q13

Which of the following methods can be used to construct the above phylogenetic relationships of the primates?

(1) comparing the fossil records of different primates

(2) comparing the DNA samples from different primates

(3) comparing the key protein samples from different primates

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

17. 2024Q15

The graph below shows the masses of the same plant species grown in two different habitats:

Which of the following can be deduced from the graph?

A. The mass of the species is affected by two genes.

B. The mass of the species is affected by environment.

C. Discontinuous variation is shown in the mass of the species.

D. The mass of the species in habitat II is more affected by environment than that in habitat I.

18. 2024Q35

Question 35 refers to some information about the interaction among three organisms: butterflies, plant species X, and ants. Butterflies lay eggs on the leaf surfaces of plant species X. When caterpillars hatch from the eggs, they feed on the leaves. Plant species X secretes a sugary solution on their leaves. Ants are attracted to feed on the sugary solution and the eggs.

It is believed that the ancestors of plant species X did not possess the structure to secrete a sugary solution on their leaves. Which of the following are the possible reasons that could have led to the emergence of this structure in plant species X?

(1) Spontaneous mutations of leaf cells of the ancestors.

(2) Gene mutation during meiotic cell division of the ancestors.

(3) Individuals with this structure survived better than those without this structure.

A. (1) and (2) only

B. (1) and (3) only

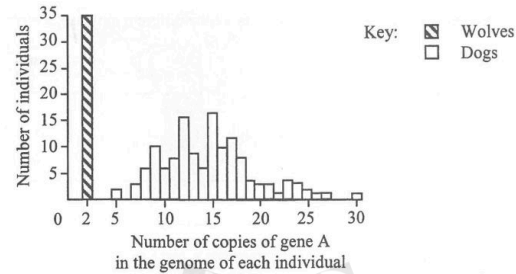
C. (2) and (3) only

D. (1), (2) and (3)

LQ

1. 2014Q6b, c

It is generally believed that domestic dogs evolved from ancient wolves. A recent study comparing the genomes of wolves and dogs suggests that genes with key roles in starch digestion were selected during the domestication of wolves into dogs. One of these genes was gene A, which codes for amylase. This gene may exist in many copies in a genome. The following graph shows the number of individuals having different numbers of copies of gene A in 35 wolves and 136 dogs.



- b) It is hypothesized that in ancient times, wolves might have been attracted to waste dumps near early human settlements and consumed human food waste. Suggest how the domestication of wolves would have led to the selection of multiple copies of gene A. (5 marks)
- c) Describe an experiment which can compare the different amylase activities of wolves and dogs. (4 marks)

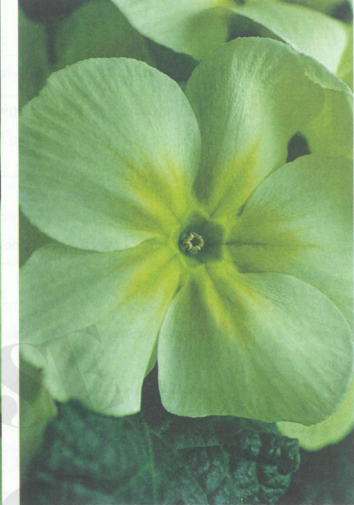
2. 2014Q8biii

A primrose plant is a flowering plant that has two different types of flowers. Photograph I and Photograph II show the appearances of the two types of flowers (P and Q) and Photograph III and Photograph IV show the sections of the flowers respectively. Each primrose plant produces either type P flowers or type Q flowers.

Photograph I: Type P flower (x 4)



Photograph II: Type Q flower (x 4)



Photograph III: Section of type P flower (x 4)



Photograph IV: Section of type Q flower (x 4)

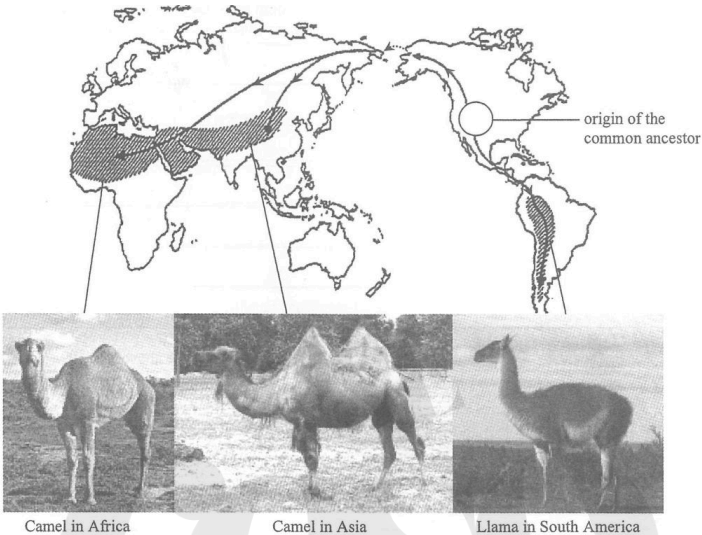


nectary

- b) Butterflies collect nectar from flowers using a mouth structure in the form of a long sucking tube. (2 marks)
- iii) What is the advantage of having the different positioning of anthers and stigmas in the primrose? (2 marks)

3. 2015Q10

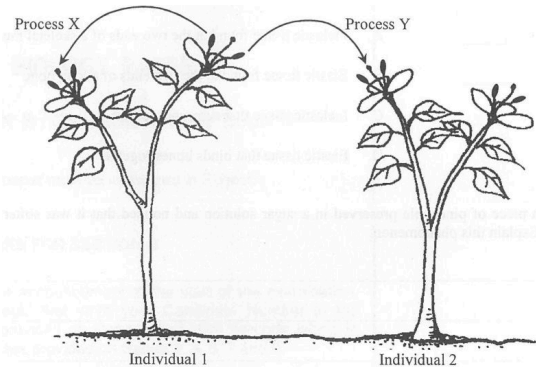
Fossil records suggest that camels in Africa and Asia and llamas in South America evolved from a common ancestor 6 million years ago. The diagram below shows the possible migration routes of the common ancestor at the time before the continents were separated and the locations (shaded areas) where the camels and llamas are found at present.



- a) Based on the information given, draw a diagram to show the evolutionary tree of the three animals. (2 marks)
- Evolutionary tree of camels in Africa and Asia and llamas in South America
- b) Explain how the common ancestor might have given rise to the two different animal species (camels and llamas) in the above case. (4 marks)
- c) Suggest another way to establish the evolutionary relationship among the above animals. (1 mark)
- d) Give **two** limitations of fossil records as evidence for evolution. (2 marks)

4. 2017Q3c

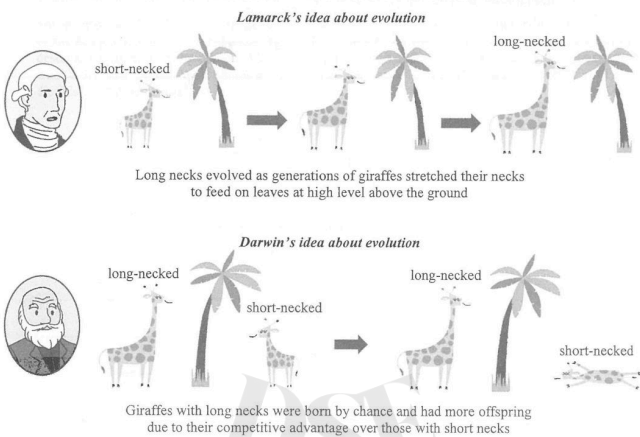
The diagram below shows certain processes occurring in a species of flowering plant:



- c) Explain briefly why process Y is better than process X in terms of evolution. (2 marks)

5. 2018Q10

Both Lamarck and Darwin contributed greatly to the understanding of evolution. They proposed different explanations for the evolution of long necks in giraffes, as shown below:



- a) Put a '✓' in the appropriate box to indicate whether the ideas listed were proposed by the scientists. You may put more than one '✓' for each idea. (2 marks)

	Lamarck	Darwin
Characters acquired during life time can be passed on to the next generation		
Organisms become more adaptive to the environment over the generations		

- b) Based on the current understanding of evolution, elaborate on the view of Darwin on the evolution of the giraffe's long neck. (4 marks)
- c) In Darwin's era, his idea caused intense debates. The concept of humans evolving from common ancestors along with other species was unacceptable to many people due to their religious belief. What does this tell us about the nature of science? (1 mark)

6. 2019Q6e
- e) In the middle of the 19th century, there was a severe attack on potato crops by a plant pathogen in Ireland. As the potato was the major staple food at that time, many Irish people died of starvation. Vegetative propagation of potatoes was blamed for the high vulnerability of the potato crops. Explain the rationale for this claim. (2 marks)

7. 2019Q8

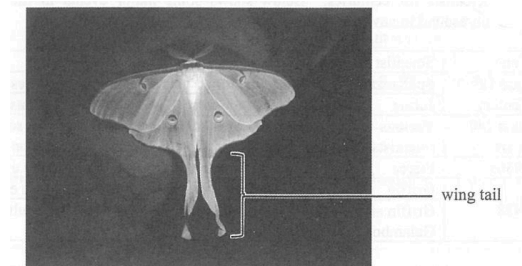
- a) Bats are night-time animals. The mystery of how bats avoid obstacles in darkness has puzzled scientists for centuries. Below shows some major events in the research on how bats use ultrasound to navigate as they fly:

Time	Scientist	Event
Late 18 th Century	Spallanzani	He noted that blind bats could avoid obstacles.
	Jurine	He plugged the ears of bats with wax; the bats collided with obstacles.
Next 140 years	Various scientists	Despite the work of Spallanzani and Jurine, scientists continued to explore the possible use of other senses for navigation in bats.
1930s	Pierce	He developed an apparatus that could detect ultrasound.
	Griffin	He used Pierce's apparatus to show that bats emitted ultrasound.
1938	Griffin and Galambos	They worked out how bats used the ultrasound they produced in navigation.

- i) With regard to the observation of Spallanzani, what conclusion can you draw about bats' ability to avoid obstacles? (1 mark)
- ii) Below are some aspects of the nature of science which can be demonstrated by the discovery of ultrasound navigation of bats. Choose any **two** aspects of nature of science and elaborate on how these are demonstrated in the above events. (2 marks)
(Note: Only the first two will be marked if you give more than two aspects.)

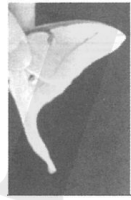
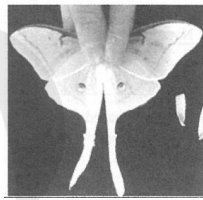
Nature of Science	Elaboration
Science is based on evidence from experiments	
Scientists build on the work of other scientists	
Technology has impacts on the development of science	

- b) Bats prey on moths. The photograph below shows a type of moth which has long wing tails:



After the discovery of ultrasound navigation in bats, scientists hypothesised that the wing tails of the moths may disturb the ultrasound emitted by bats and thus help moths to escape from a bat attack.

To test this hypothesis, scientists manipulated the wing tail length of the moths and then determined their rate of successful escape from bat attacks. The treatments of the wing tails and the results are shown below:

Treatment of wing tails	A: No treatment	B: Cut and glued back	C: Cut	D: Elongated
				
Wing tail length (cm)	5	5	2	6
Successful rate of escape (%)	57	57	26	65

- i) What can you conclude from the results of treatment A and B? What is the purpose of treatment B in the experimental design of this study? (2 marks)
- ii) What further conclusion can you draw when comparing the results of the following treatments? (2 marks)

Treatment	Conclusion
A and C	
A and D	

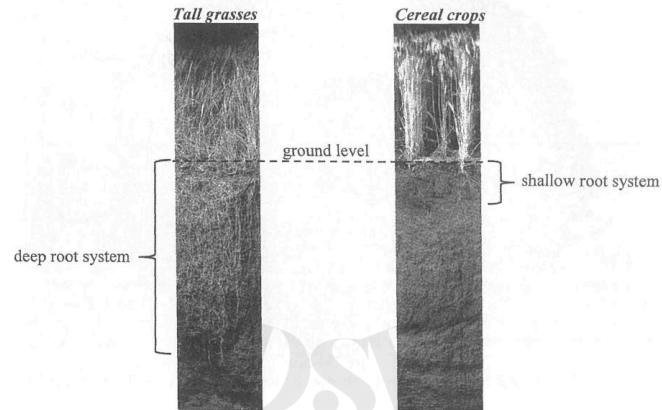
- iii) What is the overall conclusion of this study? (1 mark)
- c) With reference to the hypothesis stated in (b), describe how the long wing tail could have evolved in the moths. (4 marks)

8. 2021Q11 - Essay

Variations are important to the continuity of a population. Discuss how these variations are brought about within a population and how variations can enable the population to cope with the diverse environmental conditions and environmental changes over time. (11 marks)

9. 2022Q8a

Tall grasses and cereal crops belong to the same family and share a common ancestor. Cereal crops have been artificially selected for agriculture and their seeds harvested as food. The roots of the tall grasses in grasslands range in depth from 1.5 m to 4.5 m while those of cereal crops rarely exceed 1 m. The photographs below show the root depths of tall grasses and cereal crops under the same magnification:



- a) i) Grassland is a region of treeless plain which is mainly occupied by tall grasses. It is usually found in regions with moderate rainfall. The tall grasses have evolved with a deep root system. What is the selection pressure involved in this evolutionary process? What is the advantage of the greater root depth in tall grasses? (2 marks)
- ii) In terms of energy usage of the plant, explain why having a shallow root system for cereal crops is considered an advantage to farmers. (2 marks)

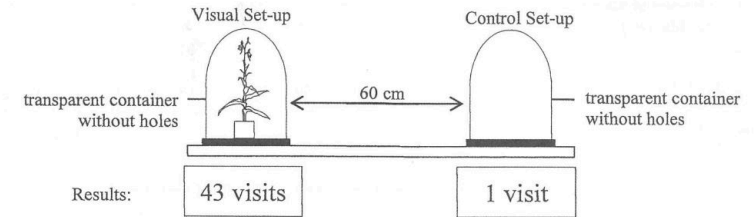
10. 2022Q10b, c

Habenaria is a genus of orchids with dull-coloured and scented flowers, which attract moths for pollination at night. In the group, *H. rhodocheila* is an exceptional species with reddish flowers which lack a detectable scent. It was found that no insects visited the flowers of *H. rhodocheila* at night for pollination while insect species A consumed nectar from the flowers in the daytime.

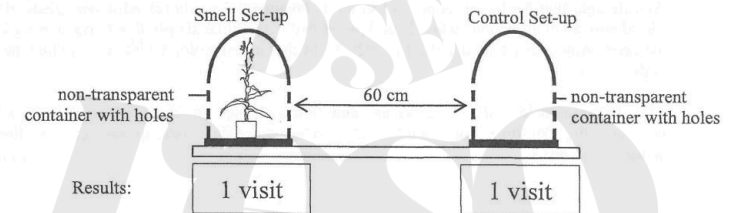
- b) In this case, *H. rhodocheila* was evolved from other *Habenaria* species. Suggest how the speciation of *H. rhodocheila* could be facilitated by different insect pollinators. (3 marks)

- c) To determine if insect species A is attracted to the flower of *H. rhodocheila* by its appearance or its smell (if any), researchers designed an investigation as shown in the diagrams below:

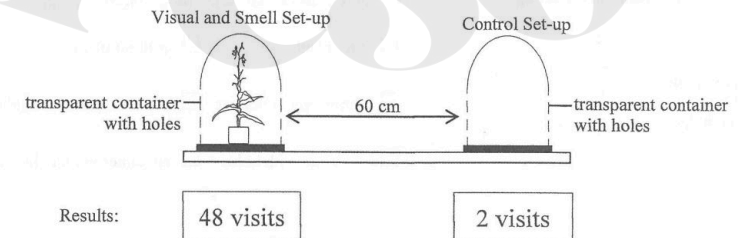
Treatment 1:



Treatment 2:



Treatment 3:



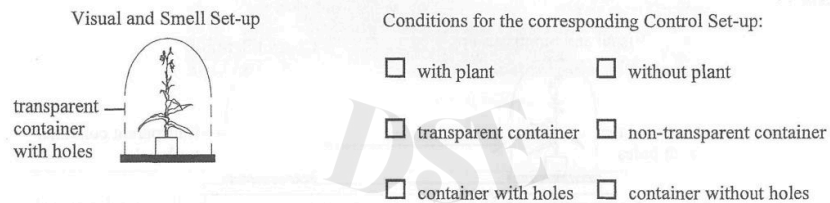
In each treatment, two set-ups were placed in an open area at a distance of 60 cm from each other for one hour. After that, the position of the two set-ups was exchanged for another hour. During the two-hour period, if an individual of species A approached any set-up to a distance of less than 10 cm, it was counted as one visit. The number of visits to each set-up is shown in the boxes in the above diagrams.

- i) To ensure that the investigation was a fair test, exchanging the positions of the two set-ups was a necessary step. Explain why. (1 mark)
- ii) With reference to the aim of the investigation, what conclusions can you draw from the results of Treatment 1 and Treatment 2 respectively? (4 marks)

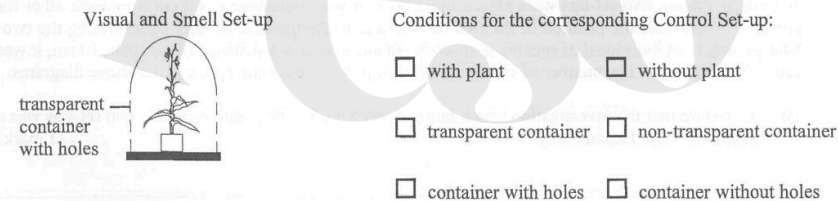
- iii) When Ann and Ken compared the results of the three treatments, they had different interpretations. Ann thought that the larger number of visits in Treatment 3 might reflect a synergistic effect of visual and smell attractions while Ken thought that it might be simply due to random variations. To verify Ann's idea, they decided to conduct a further investigation with two more treatments for comparison. (2 marks)

The diagrams below show the Visual and Smell Set-up. In each treatment, complete the corresponding Control Set-up by putting a '✓' in the appropriate boxes to show the conditions that should be adopted.

Treatment 4:



Treatment 5:



11. 2023Q11 - Essay

In agricultural practice, some crops are reproduced asexually to improve production efficiency. An increase in the yield of these crops is observed in recent years due to a steady increase in the average global temperature. Meanwhile, some scientists worry that crops reproduced asexually are at high risk of extinction due to environmental changes and diseases if global warming persists.

Explain the increased yield of these crops due to global warming and the rationale behind the concern of the scientists. (11 marks)

Ans

2013Q20: A (78%)	2013Q22: A (60%)	2014Q13: C (83%)	2014Q18: A (31%)
2016Q14: B (68%)	2017Q10: D (58%)	2017Q11: D (81%)	2020Q15: B (91%)
2020Q16: A (56%)	2020Q21: A (66%)	2022Q7: C (64%)	2023Q25: C (47%)
2023Q29: A (77%)	2024Q11: C (63%)	2024Q12: A (81%)	2024Q13: D (49%)
2024Q15: B (76%)	2024Q35: C (36%)		

2014Q6b, c

- 6 (b) • the human food waste usually contains carbohydrates / starch (1)
 • variations in copy number of gene A may exist in the ancient wolf population (1)
 • those with higher gene copy numbers could produce more amylase and were more adapted to a starch-rich diet of human food waste (1)
 • as wolves got used to feeding on human food wastes and gradually domesticated, they could then grow better & reproduce more/have more offspring than those with smaller gene copy number (1)
 • their genes, including multiple copies of gene A could be passed to the next generation (1), resulting in the selection of multiple copies in the dogs' genome (5)

- (c) Concept for mark award (4)
 • source of digestive juice (1)
 • mixing of digestive juice with starch (1)
 • appropriate test method (1), to detect either an increase in reducing sugar or a decrease in starch content
 • description and interpretation of results (1)

e.g. Measuring a decrease in starch content:

- obtain digestive juice from the mouth / intestine / pancreas of a wolf and a dog (1)
- mix 1 mL of digestive juice with 5 mL of starch solution (1)
- at regular time intervals, remove a fixed amount of the mixture for iodine test and record the time taken for the tested mixture to stay in brown colour (1)
- the shorter the time, the faster the digestion of starch, i.e. higher amylase activity (1)

Or

- obtain digestive juice from the mouth / intestine / pancreas of a wolf & a dog (1)
- drill two holes on a starch agar plate, add the digestive juices to the holes of starch agar plate (1) respectively
- incubate the plate for 30 minutes, add iodine solution to the plate and rinse, compare the brown / clear zones appeared (1)
- the one with bigger brown / clear zone shows more starch digestion, i.e. higher amylase activity (1)

Measuring an increase in the amount of reducing sugar:

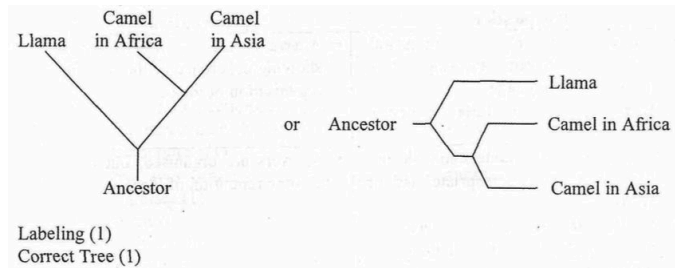
- obtain digestive juice from the mouth / intestine / pancreas of a wolf & a dog (1)
- mix 1 mL of digestive juice with 5 mL of starch solution (1)
- After 10 minutes, boil the mixture with excess Benedict's solution, and record the amount of red precipitate formed (1)
- the one with more red precipitate shows more maltose formation, i.e. a higher amylase activity (1)

2014Q8biii

- 8 (b) (iii) • this ensures that pollination is done between different individuals / avoids self-pollination (1)
 • so that the genetic variation of the offspring can be increased (1) (2)

2015Q10

10 (a)



- (b) • separation of the continents resulted in isolation of the two groups of ancestors (1)
 • each isolated group was subjected to a different set of environmental conditions (1)
 • as a result, they evolved differently from each other due to natural selection (1) / adaptive traits specific to those particular environmental conditions were selected by natural selection (1)
 • until their genetic compositions were so different that they could not interbreed again (1) (4)
- (c) • compare their genetic compositions / biochemical composition of essential proteins (1) (1)

9 marks

2017Q3c

3 (c)

- process Y involves combination of genes from two different individuals, hence the zygotes / offspring produced from process Y have greater genetic variations than those from process X (1) (2)
 • individuals with variants would be candidates for natural selection (1)

2018Q10

10 (a)

	Lamarck	Darwin
Characters developed during life time can be passed on to the ...	✓	
Organisms become more adaptive to the environment over generations	✓	✓

(2)

- (b) • genetic variation existed in the giraffe population, some with longer necks and others with shorter necks (1)
 • individuals with longer necks has better ability to obtain food / could get more food than those with shorter necks (1)
 • they have greater chance to survive and reproduce (1)
 • thus the population of the subsequent generations would have a greater proportion of giraffes with longer neck (1) (4)
- (c) • science is culturally embedded / influenced by social and cultural factors (1) (1)

7 marks

2019Q6e

6 (e)

- offspring produced will have the same genetic make-up as their parents (1)
 • this provides limited raw materials for the selection of a resistant strain / if parent genotype is susceptible to the pathogen attack, the whole population will be susceptible (1) (2)

2019Q8

8 (a)

- (i) • sight is not the sense used by bats when they avoid obstacles (1) (1)

(ii) Any **two**:

Nature of Science	Elaboration
Science is based on evidence from experiment	Spallanzani and Jurine could not provide sufficient evidence to show how bats navigated / scientists did not accept that bats use hearing for navigation until Griffin showed that bats could emit ultrasounds (1)
Scientists build on the work of other scientists	Griffin used Pierce's apparatus to show bats emitted ultrasounds / Griffin built on Spallanzani and Jurine's work, and showed that bats emitted ultrasounds, providing important evidence for the navigation (1)
Technology has impact on the development of science	Griffin could not have proved that bats emitted ultrasounds without Pierce's apparatus (1)

(2)

(accept other reasonable answers)

- (b) (i) • the procedures of cutting & gluing back the wing tail itself did not affect the rate of successful escape (1)
 • this shows that the results of C / D were related to the length of the wing tail and not related to the manipulation procedures (1) (2)

(ii)

Pair of treatments	Conclusion
A & C	• Removing / shortening the wing tail reduced the rate of successful escape (1)
A & D	• Elongating the wing tail enhanced the rate of successful escape (1)

(2)

- (iii) • the data show that the longer the wing tail, the higher the rate of successful escape (1) (1)

- (c) • within the moth population, there were genetic variations with some individuals having long wing tails and others having short wing tails (1)
 • individuals with long wing tails had a higher chance of escape from the attacks of bats than individuals with short wing tails (1)
 • therefore, individuals with long wing tails had a higher chance of survival and reproduced more offspring (1)
 • after many generations, offspring which inherited the allele for long wing tails became the majority in the population (1) (4)

12 marks

2021Q11

11 Concept for mark award:

Source of variation (max. 5):

- sexual reproduction produces offspring with a mix of genetic materials from parents (1)
 - production of gametes by meiosis (1)
 - different combinations of chromosomes due to independent assortment (1)
 - new combinations of alleles in chromosomes due to crossing over (1)
 - zygote has different combinations of genetic materials due to random fertilisation (1)
 - mutation resulting in new allele (1)
 - environmental factors, with suitable example for illustration (1)
- max. 5

Importance of variations (max 3):

- how variations cope with different environmental conditions, e.g.
 - variations lead to different niches (1), allowing the exploration of different resources / reduced competition (1)
 - how variations cope with environmental changes over time, i.e. concept of natural selection, e.g.
 - provide a range of variants within the same species for natural selection; when there are environmental changes, organisms with better adapted traits are selected and survive (1) whereas less adapted ones are eliminated (1)
- max. 3

Effective communication (0-3)

max.3
11 marks

2022Q8ai

- 8 (a) (i) • water supply is the selection force which drives the evolution of root depth / water supply in the grassland is limited (1)
- a greater root depth has a selective advantage, enabling the grasses to obtain more water from underground (1)
- (2)

2022Q10b, c

- 10 (b) Concept for mark award
- different pollinating agents → no cross breeding (1)
 - concept of isolation (1)
 - concept of speciation (1)
- (3)

e.g.

- in ancestral species of *Habenaria*, there was no cross pollination / cross breeding between individuals with dull-coloured flowers and those with reddish flowers because of the different insect pollinators (1)
- hence, the two populations were isolated and evolved independently / there was no gene flow between the two populations (1)
- after many generations, mutations / genetic variations accumulated in the two populations such that individuals of the two groups were no longer able to interbreed to produce fertile offspring (1), i.e. becoming two distinct species, the one pollinated by species A is *H. rhodocheila*

- (c) (i) to make sure data collected were not affected by the positions of the two set-ups / the preference of A was regardless of the positions of the set-ups / to cancel the effect of different physical factors (accepted named examples) of the two positions / to minimize the effect of behavioural inertia or habit of the pollinators (1)
- (1)

- (ii) Concept for mark award
- correct comparison of the results + conclusion (1+1) x 2
- (4)

e.g.

- in Treatment 1, the number of visits to the visual set-up was significantly greater than that of the control set-up (1), this shows that A was attracted by the appearance of the flower (1)
- in Treatment 2, the number of visits to the smell set-up was more or less the same as that of the control set-up (1), this shows that A was not attracted by the smell of the flower (1)

- (iii) conditions of the two control set-ups:
- with plant, transparent container without holes (1)
 - with plant, non-transparent container with holes (1)
- (2)

2023Q11

11 Increasing yields:

- an increase in temperature may speed up the enzymatic or metabolic reactions (1), this leads to an increase in the photosynthesis rate of the crops (1), there is a net production of food or an increase in the overall growth rate of the crops (1)*, leading to more food being stored in the crops or an increase in the overall size of the crops (1)[#]
 - an increase in temperature may speed up bacterial decomposition to increase the supply of inorganic minerals (1), resulting in an increase in the overall growth rate of the crops (1)*, the overall size of the crops increases (1)[#]
- max. 3

Rationale behind the scientists' worry:

- in asexual reproduction, offspring are produced through mitosis (1), so they are genetically identical to the parent plants / have no genetic variations (1)[@]
 - when there is a change in the environment which is unfavourable to the parent plants (1) all offspring will be threatened, with no exception (1)
 - micro-organisms / pathogens / pests are more active / reproduce faster at high temperatures (1)
 - all offspring will be vulnerable / susceptible to diseases or pest attack (1) as they are genetically identical (1)[@]
- max. 5

*[#]@ To be awarded once only**Effective communication (0-3)**

max. 3
11 marks